

Shivani Santhosh Kumar Bhat

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PROFESSIONAL SUMMARY

Frontend Developer and B.Tech student with hands-on experience in web development, AI-integrated applications, and IoT-based systems. Skilled in HTML, CSS, JavaScript, and React for building responsive and user-friendly web interfaces. Experienced in developing AI-powered applications using APIs and deploying them through modern web platforms. Worked on IoT-based projects using microcontrollers such as ESP32 to enable real-time data monitoring and smart device communication. Currently expanding knowledge in cybersecurity, focusing on web vulnerabilities, secure coding practices, and penetration testing concepts. Passionate about combining development, AI, and IoT expertise to build intelligent and secure digital systems.

EDUCATION

- MUTHOOT INSTITUTE OF TECHNOLOGY AND SCIENCE** September 2023 - May 2027
Full-time Bachelor of Technology Computer Science & Engineering
Kochi
CGPA: 8.15
- BHAVANS VIDYA MANDIR GIRINAGAR** May 2023 Kochi
Senior Secondary (12th Grade)
Percentage: 84.21%
- Secondary (10th Grade)** May 2021
Percentage: 79%

EXPERIENCE

- MERN FULL STACK DEVELOPER INTERN - UPTOSKILL** October 2025-February 2026
 - Actively contributing to the full-stack development life cycle of MERN applications, with responsibilities for building front-end components in React.js and creating server-side APIs using Node.js and Express.js.
 - This role also involves gaining practical experience in MongoDB database management and Git version control while assisting with debugging, code reviews, and performance optimization.

- FRONT END WEB DEVELOPMENT INTERN — EDUNET FOUNDATION** August 2025- October 2025
 - Completed a 6-week AICTE-recognised internship on Front End Web Development, implemented by Edunet Foundation in collaboration with IBM SkillsBuild.
 - Gained hands-on experience building responsive web interfaces and applying industry-aligned frontend development practices.
 - Built a Dynamic Weather Dashboard as the internship project — a real-time weather web app featuring live temperature, wind speed, air quality (PM2.5), feels-like data, and a 5-day forecast for cities across India, integrated with a live weather API and deployed on Render

SKILLS

- Security Concepts:** Web Application Security, Penetration Testing Concepts, Ethical Hacking, Vulnerability Assessment
- Programming Languages:** Python, C, Java, JavaScript
- Frameworks:** HTML, CSS, React, Angular
- Databases:** MySQL, MongoDB
- Tools & Platforms:** Git, GitHub, Render
- Soft skills:** Problem Solving, Time Management, Teamwork, Leadership, Adaptability, Conflict Resolution
- IoT:** ESP32, Arduino, Sensor Integration, Embedded Systems, Bluetooth Communication

CODING

- LeetCode:** Rank 987,059
https://leetcode.com/u/Shivani_S_Bhat/
- Hacker Rank:**
 - C programming:** Silver

PROJECTS

- CYRA

Esp32. React. JS. Machine Language

- Top 300 Finalist in All Kerala level— Tinker-her-hack
 - Developed an ESP32-based wearable device capturing real-time basal body temperature, heart rate, and motion data
 - Implemented motion artifact removal algorithms to ensure signal reliability before data reaches the ML model
 - Built an interactive React.js dashboard for real-time biometric data visualization and user-friendly cycle insights
 - Integrated a cloud backend pipeline that processes live sensor data and generates personalized cycle predictions
 - Trained machine learning models on physiological patterns, replacing calendar-based assumptions with biology-driven intelligence
 - Designed the full-stack IoT pipeline connecting hardware, cloud backend, and frontend into a seamless end-to-end system
- FIRA

ESP32 · Flutter · BLE · Arduino IDE

- Designed and developed a smart wearable exoskeleton finger rehabilitation system enabling stroke patients to perform independent home-based finger rehabilitation exercises without requiring constant physiotherapist supervision.
 - Utilised an ESP32 DevKit V1 microcontroller to drive three MG995 servo motors through a 3D-printed glove-based exoskeleton structure, producing controlled and repeatable finger flexion and extension movements via a string-spring mechanism.
 - Integrated a MAX30102 heart rate sensor and DS18B20 temperature sensor for continuous real-time physiological monitoring, with an active buzzer alert system triggered upon detection of abnormal heart rate or elevated body temperature.
 - Implemented Bluetooth Low Energy wireless communication between the ESP32 and a Flutter Android mobile application, enabling real-time transmission of servo and sensor data without internet connectivity or backend server infrastructure.
 - Developed a Flutter mobile application featuring a live session dashboard, physiological data visualisation, rehabilitation cycle tracking, and session summary reporting, distributed as a standalone offline Android APK.
- AMMAYI-GPT

React.js, Gemini-AI (gemini-2.5-flash-lite)

- Top 10 Finalist in College level — Useless Project Hackathon powered by TinkerHub
 - Built a frontend-based conversational AI web app that enables real-time, natural language interaction through a clean and responsive chat interface
 - Integrated Google Gemini API to generate context-aware, human-like responses for queries spanning explanations, brainstorming, and general assistance
 - Applied prompt engineering techniques to ensure responses remain relevant, clear, and useful across a wide range of user inputs
 - Designed a seamless conversational workflow where user input is processed in real time and responses are dynamically rendered in the chat UI
 - Built a fully responsive UI using Next.js ensuring smooth cross-device performance and accessibility
 - Deployed on Vercel for high availability and global accessibility

<https://ammayi-gpt.vercel.app>

ACHIEVEMENTS

- Selected among the Top 300 teams at the Women Makers Hackathon, Kerala, for the project CyRa — an ESP32-based wearable for biology-driven menstrual cycle prediction
- Recipient of the GHCI'26 Scholarship (Grace Hopper Celebration India 2026), awarded to outstanding women in technology builders across Kerala
- Invited to attend the Women Makers Celebration at KSUM (Kerala Startup Mission), recognizing innovative women builders across Kerala
- Reached Top 10 at college-level hackathon powered by TinkerHub for project AmmayiGPT a frontend AI chatbot powered by Google Gemini API

CERTIFICATIONS

- Ethical Hacking and Penetration Testing | L&T Learning
 - Deloitte Australia Cyber Job Simulation | Forage
 - Skyscanner Front-End Software Engineering virtual experience program | Forage
 - Google Cloud Generative AI Fundamentals | Google Cloud & Hack2Skill
- September 2025

July 2025

July 2025

July 2025